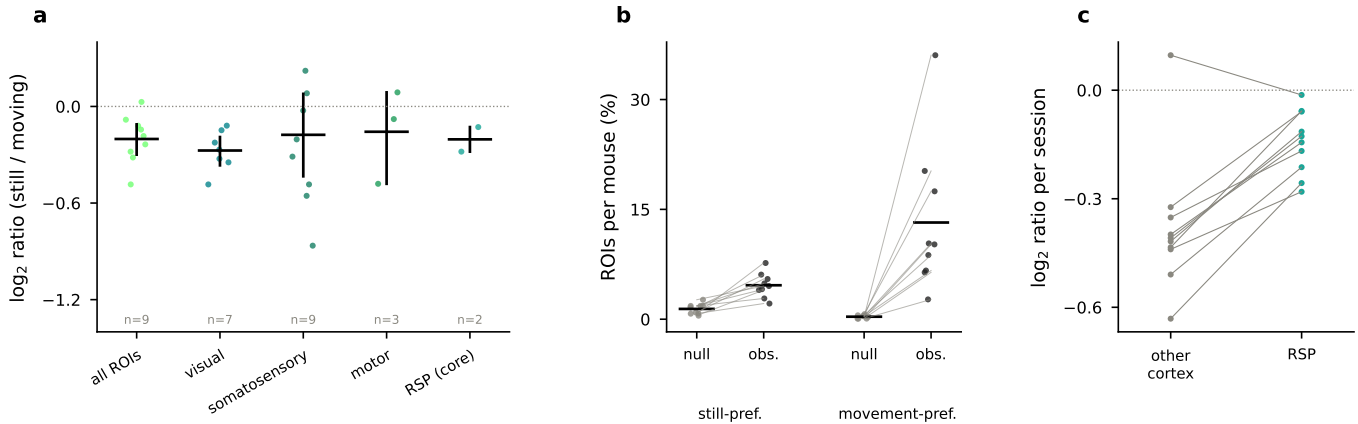

Figure 17: Correlates of stillness in the IBL mesoscope: lower activity in stillness across dorsal cortex. **a** Activity in still against moving 1 s bins of the passive block before the task replay (grey screen and receptive-field mapping; median 6.2 min per session), $\log_2((\text{still}+e)/(\text{moving}+e))$ of each ROI's deconvolved activity with e = 2% of its mean; ROIs averaged per session (sessions with $\geq$20 ROIs in the group), sessions per mouse. Moving: lick or face motion energy above the session's 60th percentile (the wheel is not recorded in this block); still: no movement within $\pm$1 s and $\leq$30 s after the last movement. Groups by Beryl region (Allen colours); black, mean and 95% bootstrap CI over mice; n, mice. **b** Still- and movement-preferring ROIs per mouse (Welch test of still against moving bins, at most one of each per 2 s, $p < 0.01$), observed against a circular-shift null (activity rotated against the behavioural labels, 100 offsets $\geq$60 s). Bars: means over mice. One session whose null exceeded 5% (slow drift; a quarter of its ROIs silent in all still bins) is excluded from all panels. **c** RSP against the other imaged ROIs of the same session (lines: sessions).